B - Material Engineering MS4092

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Number of problems: 8
Start at: 15:50 PM
Finish at: 16:12 PM
Duration: 22 Minutes

What is the main reason that white cast iron is rarely used as material for industrial tools?

sangat brittle

- ○ It needs laser welding to join the material
- ⊗ It is difficult to be deformed and cut
- ○ Its color is whittish
- ○ It needs chemical solution for cleaning

Ture strain of a tensile test specimen is 0.15 at true stress of 600 MPa. The K-constant value of its true stress-true strain relation is 825 MPa. What is the specimen strain-hardening coefficient?

- ○ 0.45
- ⊗ 0.17
- ○ 0.14
- ○ 0.31

$$\sigma_T = k \cdot (\varepsilon_T)^n$$

$$600 = 825 \cdot 0{,}15^n$$

$$0{,}15^n = \frac{600}{825}$$

$$n = \frac{\log(600/825)}{\log 0{,}15} = 0{,}17$$

Among these statements regarding low carbon steel, which one is NOT accurate?

- ○ All statements are correct
- ○ Low carbon steel has lower modulus of elasticity than that of high carbon steel
- ⊗ In general, low carbon steel is tougher than high carbon steel
- ○ Low carbon steel is easier to be corroded than high carbon steel

A microstructure of a steel consists of 60% pearlite and 40% pro-eutectoid ferrite. What is the carbon content of the steel?

- ○ 0.51%
- ○ 0.39%
- ○ 0.32%
- ⊗ 0.46%

$$a = \frac{b - X}{b - c}$$

$$0{,}4 = \frac{0{,}77 - X}{0{,}77 - 0{,}0218}$$

$$X = 0{,}47\%$$

What is the main reason that a stainless steel has a good corrosion resistance ?

- ○ It cannot rust in any kind of media
- ✗ It can form a passive protective layer → from chromium reacts with O_2 in the air
- ○ By its nature it does not rust
- ○ It has high purity

Among these statements regarding the AISI 1030, which is the most accurate statement?

* high carbon steel
* Carbon content 0,3%

- ○ The steel is a low carbon steel
- ○ Its carbon content is 30%
- ✗ The other statements are wrong
- ○ Its carbon content is 3%

A carbon steel has 0.2% carbon content; what is percentage of pearlite in the carbon steel?

$$\% = \frac{0,2 - 0.008}{0,8 - 0,008} \times 100\% = 24,2\%$$

- ○ 75.9%
- ○ 44.1%
- ✗ 24.1%
- ○ 55.9%

A cylindrical carbon steel rod has diameter of 12.5 mm and yield strength of 345 MPa. Which is the correct statement related to the steel rod?

- ○ It needs at least about 42 kN force to break the rod
- ○ The highest possible elongation of the rod is about 32.5%
- ✗ A force of about 42.3 kN or higher is needed to permanently deform the rod
- ○ It needs minimum force of about 35.7 kN to deform the rod plastically

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